

Effects of Salinity on Growth, Metabolism, and Ion Regulation in Juvenile Rainbow and Steelhead Trout (*Oncorhynchus mykiss*) and Fall Chinook Salmon (*Oncorhynchus tshawytscha*)

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Rainbow and steelhead trout (*Oncorhynchus mykiss*) and fall chinook salmon (*Oncorhynchus tshawytscha*) fry were acclimated to a range of salinities to test the hypothesis that the energy requirements for ion regulation would be minimal at an isotonic water salinity (8–10 ppt). Survival, growth, metabolic rate, plasma Na^+ and Cl^- concentrations, and seawater adaptability were measured for 5–12 wk, depending on the species. Growth of all three species was highest in fresh water and declined with increasing salinity. Metabolic rates increased with salinity and were inversely correlated with growth rates. Isotonic salinity, therefore, did not offer significant metabolic or growth advantages to rainbow, steelhead, and chinook fry. While plasma Na^+ and Cl^- concentrations varied among groups, chinook fry tended to better maintain ionic homeostasis at higher salinities than the trout. Acclimation to the various dilute salinities did not influence the seawater adaptability of juvenile steelhead trout or chinook salmon. Our results indicate that optimal salinities for growth and metabolic rates were influenced by species and life history stage. The metabolic rate data suggested that the energetic cost of ionic regulation increased with salinity, but attempts to quantify this cost were probably affected by other metabolic processes which responded to changes in salinity.

On a acclimaté des alevins de truite arc-en-ciel et de truite arc-en-ciel anadrome (*Oncorhynchus mykiss*) et de saumon quinnat (*Oncorhynchus tshawytscha*) à diverses salinités afin de vérifier l'hypothèse selon laquelle les besoins énergétiques de la régulation ionique seraient faibles à une salinité isotonique (8–10 ppm). Puis on a quantifié le taux de survie, le taux de croissance, le rythme métabolique, les teneurs en Na^+ et en Cl^- du plasma ainsi que le niveau d'adaptation au milieu marin pendant 5 à 12 sem, selon l'espèce. Le taux de croissance des trois espèces était plus élevé en eau douce et a diminué en fonction d'une augmentation de la salinité. Par contre, leur rythme métabolique a augmenté en fonction de la salinité et était en corrélation inverse avec leur taux de croissance. La salinité isotonique n'a toutefois pas offert d'avantages significatifs au niveau métabolique. Quoique les teneurs en Na^+ et en Cl^- dans le plasma variaient d'un groupe à l'autre, les alevins de saumon quinnat avaient tendance à maintenir une meilleure homéostasie ionique à des salinités élevées que les alevins de truite. L'acclimation aux diverses salinités diluées n'a pas influé sur la capacité d'adaptation des juvéniles de truite arc-en-ciel anadrome et de saumon quinnat au milieu marin. Les résultats de l'étude révèlent que l'espèce et le stade du cycle vital influent sur les salinités optimales en ce qui concerne la croissance et le métabolisme. Les données pertinentes au rythme métabolique portent à croire que le coût énergétique de la régulation ionique augmente en fonction de la salinité, mais les essais de quantification de ce coût ont probablement été affectés par d'autres processus métaboliques qui ont réagi aux variations de la salinité.

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There are conflicting reports in the literature regarding the effects of isotonic salinity (i.e. 8–10 ppt; Brett 1979) on the growth and metabolic rate of salmonids. Both Canagaratnam (1959) and Otto (1971) found that growth rates of presmolt coho salmon (*Oncorhynchus kisutch*) were higher between 5 and 12 ppt than in freshwater. Measurements of oxygen consumption in rainbow trout (*Oncorhynchus mykiss*) further showed a reduction in metabolic rate (20–28%) in isotonic salinity relative to fresh water and seawater (Rao 1968). These studies support the hypothesis that the energetic cost of ion regulation is lowest in an isotonic environment, where the ionic gradients between blood and water are minimal, and that this energy savings is substantial enough to increase growth. Other studies, however, fail to show isotonic salinity as the point of

maximal growth in salmonids; growth rates were highest in fresh water in studies by Shaw et al. (1975), Clarke et al. (1981), McKay and Gjerde (1985), and McCormick et al. (1989c). A theoretical estimate of the energetic cost of osmoregulation in salmonids suggests that it would be very low, being less than 1% of resting metabolism (Eddy 1982). Accordingly, changes in metabolic rate of that magnitude would be very difficult to measure accurately. Studies such as that of Bullivant (1961), who reported no significant differences between the metabolic rates of yearling chinook salmon (*Oncorhynchus tshawytscha*) reared in fresh water, half-strength seawater, and full-strength seawater, support those theoretical estimates.

Few studies have made simultaneous measurements on growth and metabolic rates of salmonids in relation to salinity.

To study those relationships, we examined growth, metabolic rates, and ionic regulation in juvenile rainbow and steelhead trout and fall chinook salmon acclimated to a range of salinities, with one near isotonic. These species were chosen to investigate the possible effects of different life history patterns on growth and metabolic responses to environmental salinity. The steelhead trout is an anadromous form of the rainbow trout, usually spending 2 yr in fresh water before migrating to the sea (Scott and Crossman 1973). Fall chinook salmon are also anadromous, but spend a shorter period of time in fresh water than steelhead trout, migrating seaward within 3 mo of emergence (Lister and Walker 1966).

Materials and Methods

Fish Stocks

Domesticated rainbow trout fry were obtained from Sun Valley Trout Farms, Mission, B.C., in November 1988. Steelhead trout and fall chinook salmon eggs were taken at the Big Qualicum River hatchery on Vancouver Island in April and November 1989, respectively, and incubated to the swim-up stage at the Pacific Biological Station in Nanaimo. All three stocks were transferred to the E.V.S. Consultants Marine Laboratory in North Vancouver and were held in fibreglass tanks with running fresh water for up to 3 wk prior to experimentation.

Growth Experiments

The experimental design for each growth trial is shown in Table 1. At the start of the experiments, the fry were lightly anesthetized in 2-phenoxyethanol ($0.5 \text{ mL} \cdot \text{L}^{-1}$), measured for length (to the nearest millimetre) and weight (to the nearest 0.01 g), and added to 180-L fibreglass tanks supplied with both fresh water and seawater. One tank was prepared for each salinity treatment. Flow rates of fresh and salt water into common header mixing buckets were controlled by valves to achieve the desired salinities (± 1 ppt) in the rearing tanks. The fish were gradually acclimated to the test salinities in a stepwise manner at a rate of $1\text{--}2 \text{ ppt} \cdot \text{d}^{-1}$. Salinity, temperature, dissolved oxygen, pH, and mortalities were recorded daily in each rearing tank. Water temperatures were kept constant ($\pm 1^\circ\text{C}$) using immersion heaters, and aeration was supplied to each tank to maintain dissolved oxygen levels near saturation. Dissolved oxygen levels ranged from 8.6 to $10.7 \text{ mg} \cdot \text{L}^{-1}$ due to differences in oxygen solubility with increasing salt content, but the minimum concentrations were not considered to be limiting to

growth (Brett 1979). The fish were fed 4–5% of their total body weight daily by hand, using a diet of dry EWOS fry feed. The fish actively fed and there were no observable differences in swimming activity between the different salinity treatments.

Random samples of 20 steelhead trout and chinook salmon fry per treatment were measured for length and weight after the acclimation periods, and at 3-wk intervals throughout the experiments, to determine growth rates and feeding rations. Percent moisture content was determined from a sample of 10 chinook fry per treatment at each sampling period after oven drying at 80°C for 24 h. Blood was also collected from 10 fish per treatment at several sampling periods for the determination of hematocrit values and ion concentrations. After the fish was anesthetized, the caudal peduncle was severed and blood was collected from the dorsal aorta into heparinized capillary tubes. Hematocrits (percent red blood cells by volume) were determined after centrifugation at 10 000 rpm for 5 min. The separated plasma was withdrawn using Microcap® tubes and frozen at -20°C in 0.5-mL plastic vials prior to ion concentration analyses. Because of low plasma volumes in the smaller fish, it was necessary to pool samples in some cases to obtain sufficient volume for the ion analyses. Plasma chloride concentration ($[\text{Cl}^-]_{\text{pl}}$) determinations were made by coulometric titration (Haake Buchler Instruments digital chloridometer), while plasma sodium concentrations ($[\text{Na}^+]_{\text{pl}}$) were measured by flame photometry (Corning flame photometer model 410). Appropriate reference standards were run every 10 samples to ensure measurement accuracy. Water was collected from each rearing tank midway through the growth trials and analysed for Na^+ , Cl^- , and Ca^{2+} for comparison with blood plasma values (Table 2).

At the end of each growth experiment, all remaining fish were measured for length and weight and 10 fish from each treatment were killed for hematocrit, $[\text{Na}^+]_{\text{pl}}$, and $[\text{Cl}^-]_{\text{pl}}$ determinations. Specific growth rates (expressed as percent change per day) were calculated for each treatment as $\ln(\text{Wt}_f/\text{Wt}_i) \cdot \text{d}^{-1} \cdot 100$ where Wt_i and Wt_f are the initial (after salinity acclimation) and final mean weights, respectively. Condition factors ($K = \text{wt} \cdot \text{L}^{-3} \cdot 100$; Ricker 1975) were also calculated from the length and weight data.

Respirometry Experiments

After the growth experiments were completed, oxygen consumption rates were measured for fish acclimated to the test salinities. The respirometers consisted of 1-L cylindrical glass chambers, 12 cm in diameter and 9 cm in height, and designed

TABLE 1. Experimental conditions for juvenile salmonid growth trials in different water salinities.

Parameter	Rainbow trout	Steelhead trout	Chinook salmon
Timing	Nov.–Dec. 1988	June–Sept. 1989	Feb.–June 1990
Duration	5 wk	12 wk	11 wk
Salinities (ppt)	0, 9 ^a , 18	0, 4, 8 ^a , 12, 16	0, 5, 10 ^a , 20, 28
Temperature ($^\circ\text{C}$)	11	14	11.5
Photoperiod (h)	12 (constant)	14 (constant)	11–15 (natural)
No. of fish per tank	50	80	100
Initial mean weights (g)	1.0	0.4	1.0
Sampling frequency	5 wk	3 wk	3 wk

^aClosest to isotonic value (i.e. 8–10 ppt).

TABLE 2. Chemical composition of water samples collected from the steelhead trout and chinook salmon fry rearing tanks (FW = freshwater; SW = seawater).

Salinity (ppt)	Chemical parameter			
	Na ⁺ (meq·L ⁻¹)	Cl ⁻ (meq·L ⁻¹)	Ca ²⁺ (meq·L ⁻¹)	pH
<i>Steelhead trout</i>				
0 (FW)	3	<1	<1	6.0
4	71	63	1	7.1
8	146	120	1	7.3
12	189	170	2	7.4
16	213	240	3	7.5
28 (SW)	386	450	6	7.7
<i>Chinook salmon</i>				
0	3	<1	<1	6.0
5	81	71	1	7.1
10	163	156	3	7.3
20	273	288	5	7.5
28	371	456	7	7.6

to sit on magnetic stir plates. Water from an aerated and temperature controlled reservoir was passed through the chambers using a Masterflex® pump. Thick-walled Nalgene® tubing was used to connect the chambers to dissolved oxygen probes and a four-way valve assembly which allowed the respirometer to operate in an open (flow-through) or closed mode. Circular water currents within each respirometer chamber were generated by magnetic stirrers and speeds were controlled by calibrating digital rpm readouts on the stir plates with velocities determined for each chamber. Dissolved oxygen levels in each chamber were monitored continuously using polarographic oxygen electrodes and a multichannel oxygen meter (Orbisphere model 2710).

Measurements of respiratory metabolism in fish can be influenced by several factors including water temperature and fish size and activity (Brett and Groves 1979). To determine the effect of salinity alone, concerted efforts were made to control all other variables and therefore minimize variation in test results. Depending on the species, two to five fish of similar size were placed into each respirometer through a ground-glass lid and the flow rate was adjusted to approximately 100 mL·min⁻¹. The fish were acclimated to the respirometers for 24 h and were not fed for 48 h prior to testing to ensure a postabsorptive digestive state (Brett 1964). To shield the fish from outside disturbances, the respirometers were covered with black plastic throughout acclimation and testing. After acclimation, the fish were given a 15-min swimming adjustment period before the respiration trial. In each trial, the swimming speed was set to 1 body length·s⁻¹ to standardize the level of activity (Brett 1962). This mild swimming challenge was necessary to reduce the variation in oxygen consumption caused by spontaneous activity at swimming speeds less than 1 body length·s⁻¹ (Brett and Glass 1973). The respirometers were then closed and the subsequent decline in water oxygen concentration (to the nearest 0.05 mg·L⁻¹) was monitored for 1 h, with values recorded every 5 min. After a trial, the fish were removed and measured for length and weight. The trials were conducted at approximately the same time each day to minimize diurnal variation in metabolism due to entrainment to a feeding schedule or photoperiod (Brett and Zala 1975). Water temper-

atures were kept similar to those used in the growth experiments (Table 1). Background oxygen consumption, due to bacteria, was measured by running blanks (i.e. no fish) throughout the experimental period. A correction was applied to the fish respiration measurements assuming a linear increase of bacterial oxygen consumption.

Water oxygen concentrations initially declined in a curvilinear fashion for about 10–15 min, apparently as a result of pressure changes associated with closing the system (Rombough 1988). Oxygen levels then decreased at a constant rate for about 30–45 min. Approximately 25% of the initial oxygen was consumed during the linear portion of the measurements. Oxygen consumption rates were estimated over the linear portion using regression analysis. Metabolic rates were determined by multiplying the regression slope of oxygen uptake by the respirometer volume and were expressed in terms of wet weight over a 1-h period (i.e. milligrams oxygen per kilogram per hour). Six replicate trials were conducted per salinity treatment for the steelhead trout and chinook salmon and three for the rainbow trout fry.

Preliminary trials with rainbow trout fry in fresh water showed that oxygen consumption increased linearly with increasing fish weight when expressed on a double logarithmic grid (slope = 0.77). This is a well-established relationship in studies on respiratory metabolism of fish (Brett and Groves 1979) and was used in the present study to validate the method of measuring oxygen consumption in the circular chambers.

Seawater Challenge Testing

At the end of the study, juvenile steelhead trout and chinook salmon (average weights = 7–10 g) were subjected to seawater challenge tests (Blackburn and Clarke 1987) to determine the effect of prior acclimation to the dilute salinities on subsequent adaptability to full-strength seawater. Random samples of 10–12 fish were transferred from the test salinities to 28–29 ppt (with the exception of the chinook 28 ppt treatment which had only four fish of small size remaining) and were sampled after 24 h of exposure for [Na⁺]_{pl} as described before. The criterion for proper seawater adaptation is the ability to maintain [Na⁺]_{pl} less than 170 meq·L⁻¹ 1 d after transfer to seawater (Clarke et al. 1981).

Statistical Analysis

Data are presented as means ± standard error (SE) where appropriate. When analysis of variance indicated significant effects, Tukey's multiple comparison test (Steel and Torrie 1980) was used to identify significantly different treatment means ($P < 0.05$). All analyses were performed with the SYSTAT statistical program (Wilkinson 1988).

Results

Mortality

Mortality was less than 5% in all treatments except for chinook salmon fry reared in 28 ppt, which suffered a loss of 24 fish (Table 3). The majority of the mortalities were less than 5 cm in fork length and weighed less than 1.5 g.

Growth and Condition

Growth rates declined with increasing salinity for all three species (Fig. 1; Table 3). Rainbow trout fry growth decreased

TABLE 3. Mortality, specific growth rates, condition factors, and moisture content of juvenile rainbow and steelhead trout and chinook salmon reared in various salinities. Mean values followed by the same letter are not significantly different ($P < 0.05$) by Tukey's test.

Salinity treatment (ppt)	Percent mortality	Specific growth rate (% wt·d ⁻¹)	Final mean condition factor (± SE)	Final mean percent moisture content (± SE)
<i>Rainbow trout</i>				
0	2	3.35	1.13 ± 0.01 a	
9	0	3.25	1.17 ± 0.01 a	
18	4	1.57	1.07 ± 0.01 b	
<i>Steelhead trout</i>				
0	3	3.27	1.12 ± 0.01 a	
4	0	2.67	1.15 ± 0.01 b	
8	0	2.98	1.19 ± 0.01 c	
12	0	3.11	1.17 ± 0.01 bc	
16	0	2.75	1.17 ± 0.01 bc	
<i>Chinook salmon</i>				
0	0	2.62	1.05 ± 0.01 a	78.0 ± 0.2 a
5	0	2.44	1.06 ± 0.01 a	78.1 ± 0.2 ab
10	0	2.32	1.06 ± 0.01 a	78.6 ± 0.2 ab
20	0	2.30	1.07 ± 0.01 a	79.1 ± 0.2 b
28	24	1.84	1.15 ± 0.01 b	81.4 ± 0.4 c

slightly from fresh water to isotonic salinity and decreased significantly in the hypertonic treatment. Steelhead trout fry showed a similar response, but final weights in 4 ppt were significantly lower than in fresh water (Fig. 2). Growth of chinook salmon fry was significantly higher in fresh water than all other treatments and showed a steady decline with increasing salinity to full-strength seawater. Specific growth rate values for chinook salmon fry in fresh water and seawater were similar to those reported by Clarke et al. (1981).

Condition factors of steelhead trout and chinook salmon fry did not differ significantly between salinity treatments for the first three sampling periods of the growth trials. Final condition factors varied among growth experiments, showing no consistent pattern between species (Table 3). Rainbow trout fry in fresh water and isotonic salinity had significantly higher final condition factors than the hypertonic treatment, but the condition factor for steelhead trout fry in fresh water was significantly lower than the other treatments. Condition factors for the chinook salmon fry were similar up to 20 ppt, but were significantly higher in 28 ppt; fish in full-strength seawater tended to be deeper bodied than the more slender body forms in the other treatments. The initial moisture content of the chinook salmon fry was 81% in all treatments and, as expected, declined with growth of the fry (Clarke et al. 1981). Differential growth rates between salinity treatments, however, resulted in size differences which masked the anticipated dehydration effect of salinity on moisture content (Table 3).

Hematocrit and Plasma Ion Concentrations

Hematocrit values for rainbow trout fry increased with salinity and were significantly higher in 18 ppt than in fresh water (Fig. 3). Hematocrit values for steelhead trout fry were also significantly higher in 16 ppt than in fresh water on day 63, but this pattern was not repeated at the other sampling times

(Fig. 4). Hematocrit values for chinook salmon fry followed an opposite trend from the trout and decreased with increasing salinity, being significantly lower in 20 and 28 ppt than in fresh water on day 56 (Fig. 5). $[Na^+]_{pl}$ measured in rainbow trout did not differ significantly among treatments, while $[Cl^-]_{pl}$ was significantly higher in 18 ppt (Fig. 3). $[Na^+]_{pl}$ in steelhead trout fry tended to increase with salinity and was significantly higher in 12 and 16 ppt compared with fresh water on day 84. $[Cl^-]_{pl}$ did not differ significantly among treatments, although it tended to be higher at 12 and 16 ppt on days 42 and 63 (Fig. 4). $[Na^+]_{pl}$ and $[Cl^-]_{pl}$ in chinook salmon fry did not differ significantly among treatments at each sampling period (Fig. 5).

Metabolic Rates

The average metabolic rates of juvenile salmonids in fresh water measured in the present study ranged from 207 to 270 mg O₂ kg⁻¹·h⁻¹, depending on species and temperature, and were comparable with values obtained by Brett (1965) for juvenile sockeye salmon (*Oncorhynchus nerka*) of similar size and activity. Metabolic rates of all three species increased with salinity, being significantly higher in hypertonic treatments than in fresh water (Fig. 6). The magnitude of these changes in metabolic rate from fresh water to 16–20 ppt ranged from 25 to 30%. The following linear regression equations describe the relationship between metabolic rate and salinity, determined for the three species:

- (1) Rainbow trout: $\ln \dot{M}O_2 = 0.080 \cdot \ln (S + 1) + 5.328$;
 $r^2 = 98.6\%$
- (2) Steelhead trout: $\ln \dot{M}O_2 = 0.064 \cdot \ln (S + 1) + 5.581$;
 $r^2 = 75.2\%$
- (3) Chinook salmon: $\ln \dot{M}O_2 = 0.087 \cdot \ln (S + 1) + 5.569$;
 $r^2 = 93.7\%$

where $\dot{M}O_2$ is metabolic rate (milligrams oxygen per kilogram

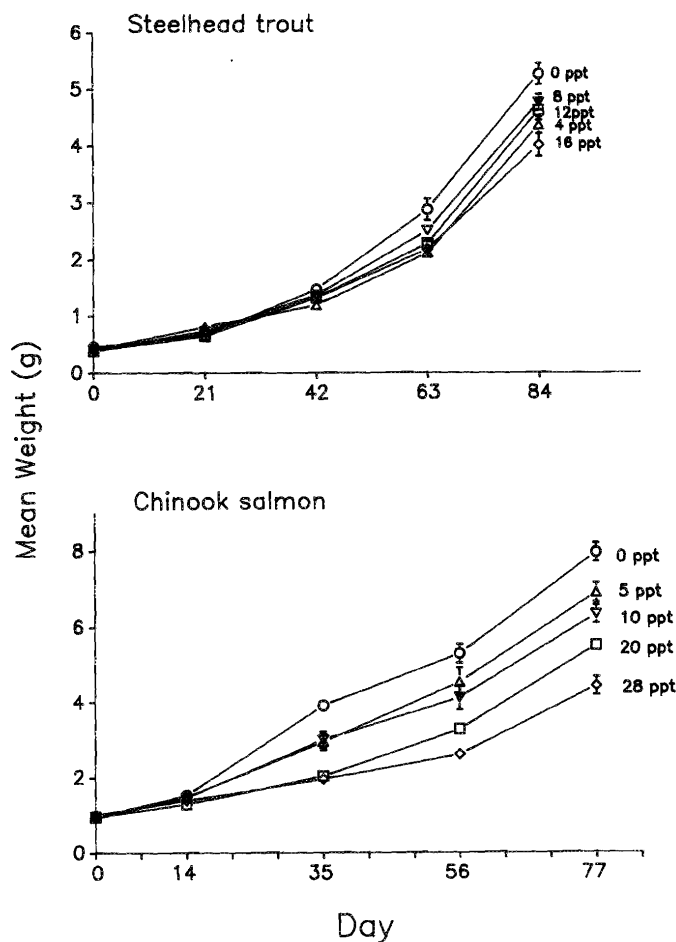


FIG. 1. Growth curves for steelhead trout and chinook salmon fry reared in five different salinities. Absence of standard error bars indicates that the SE was smaller than the symbol.

per hour) and S is salinity (parts per thousand). These equations provide good estimates of metabolic rates for the size of fish, water temperatures, activity level, and range of salinities used in this study. The increases in metabolic rate also showed significant inverse linear correlation to the observed decreases in growth (r^2 values: rainbow = 0.69, steelhead = 0.63, chinook = 0.89).

Seawater Challenge Tests

No mortalities were recorded during the 24-h seawater challenge tests. $[Na^+]_{pl}$ in juvenile steelhead trout and chinook salmon after 24 h of exposure to seawater were all below the $170 \text{ meq}\cdot\text{L}^{-1}$ threshold and did not differ significantly among treatments, with the exception of steelhead trout transferred from 4 ppt which averaged $177.4 \pm 3.6 \text{ meq}\cdot\text{L}^{-1}$ (Fig. 7). Acclimation to dilute salinities therefore had little influence on the seawater adaptability of juvenile steelhead trout and chinook salmon in the present study.

Discussion

The mortality data indicate that the small fry stages (approximately 0.5–1.0 g) of these species can adapt to salinities of up to 20 ppt when a gradual acclimation procedure is followed. Previous studies on seawater adaptation have also

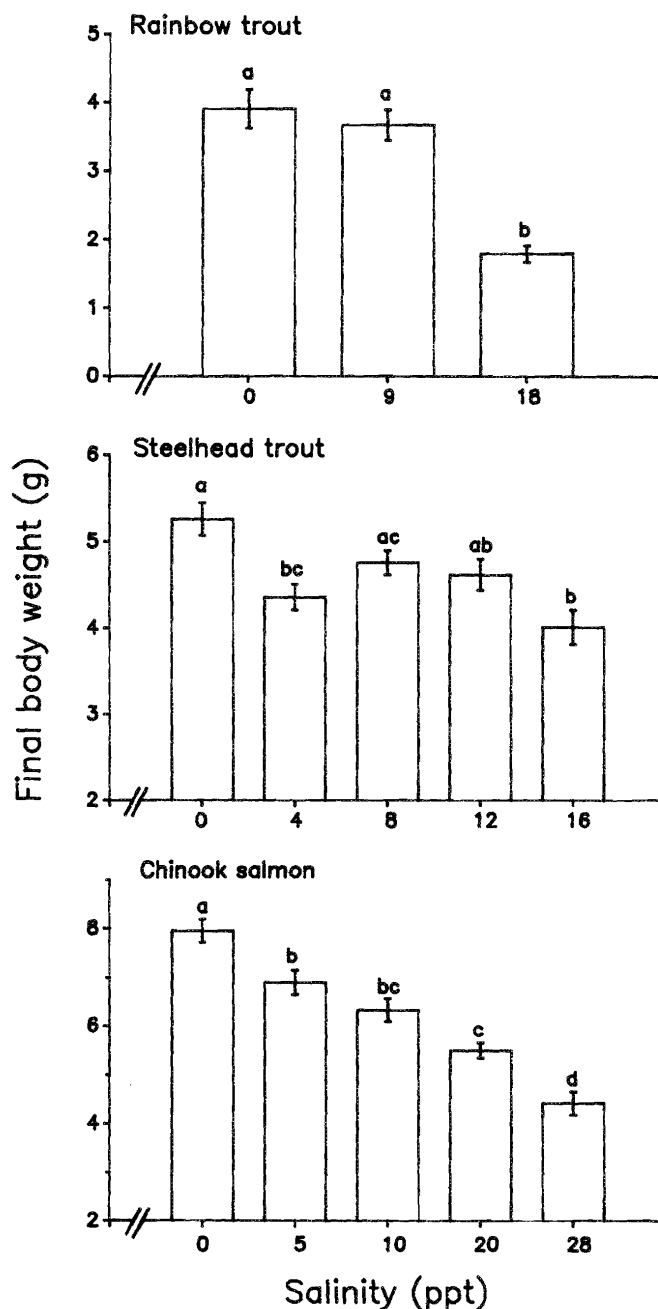


FIG. 2. Effect of salinity on growth of juvenile rainbow and steelhead trout and chinook salmon (starting weights = 1.0, 0.4, and 1.0 g, respectively). Means (± 1 SE) with a common letter are not significantly different by Tukey's test.

shown increased survival of 1-g chinook salmon fry following gradual transition to seawater (Wagner et al. 1969; Kepshire and McNeil 1972) and 20 ppt salinity to be an upper threshold limit for the survival of slightly larger (10–30 g) rainbow trout (Landless 1976; Eddy and Bath 1979; Jackson 1981; Johnsson and Clarke 1988) and Atlantic salmon (*Salmo salar*) parr (Saunders and Henderson 1969a).

The measured values for both $[Na^+]_{pl}$ and $[Cl^-]_{pl}$ in fresh water are in good agreement with normal values for all three species (Conte and Wagner 1965; Wagner et al. 1969; Hille 1982). $[Na^+]_{pl}$ and $[Cl^-]_{pl}$ in rainbow and steelhead trout rose slightly when reared in salinities above the isotonic level, while

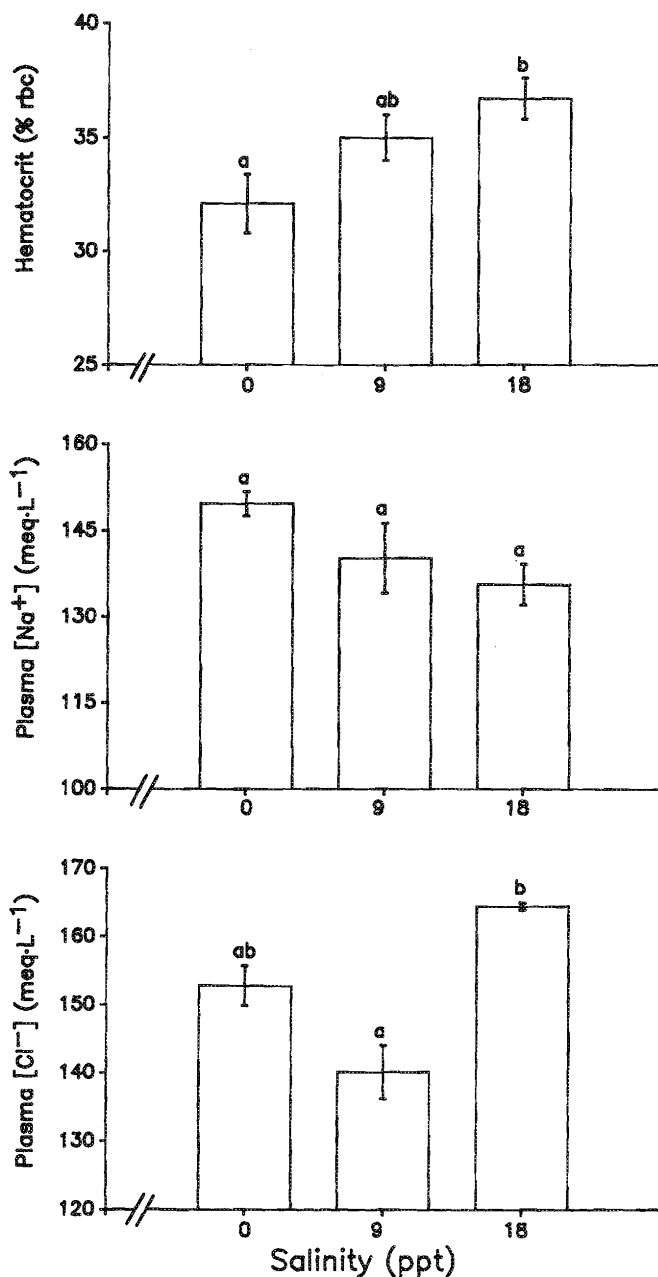


FIG. 3. Mean hematocrit and $[Na^+]_{pl}$ and $[Cl^-]_{pl}$ for rainbow trout fry after 5 wk in three different salinities. Means (± 1 SE) with a common letter are not significantly different by Tukey's test.

plasma ion concentrations in chinook salmon fry did not change across all salinities (the ionic status of mortalities in 28 ppt could not be determined). Despite these slight increases (8–10%) in the trout, plasma ion concentrations for all three species were maintained within acceptable levels (i.e. $[Na^+]_{pl} < 180 \text{ meq} \cdot L^{-1}$, $[Cl^-]_{pl} < 160 \text{ meq} \cdot L^{-1}$, Eddy and Bath 1979; Johnston and Cheverie 1985; Stagg et al. 1989), indicating that the fish were regulating blood ions competently against concentration gradients (Table 2). It is likely, therefore, that the observed growth and metabolic rates reflect steady state in fish with respect to ionic status. The increase in Ca^{2+} content of the external media at higher salinities may have assisted in regulating the passive movements of Na^+ and Cl^- , as Ca^{2+} is known to alter the permeability of gill membranes to these ions

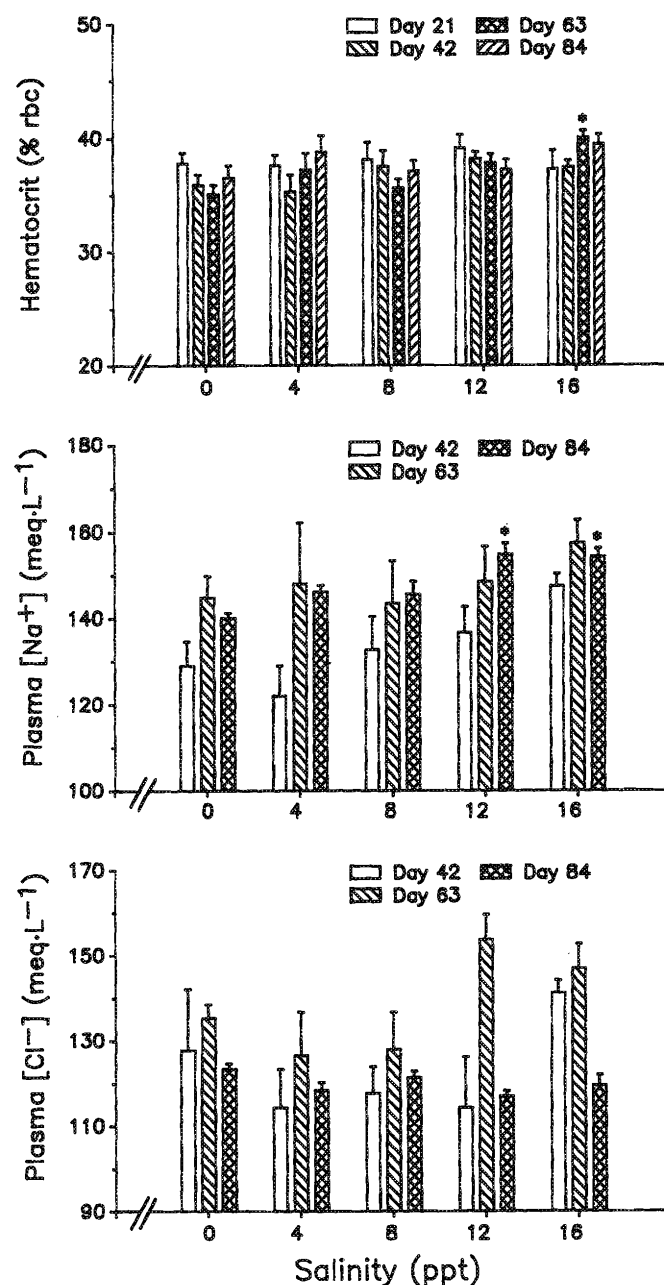


FIG. 4. Mean hematocrit and $[Na^+]_{pl}$ and $[Cl^-]_{pl}$ (+1 SE) for steelhead trout fry reared in five different salinities. Asterisks denote salinity treatment means which are significantly different from the freshwater value.

(Eddy 1975). The variable responses of hematocrit to salinity observed in the present study have been reported previously (Leatherland and McKeown 1974; Zeitoun et al. 1974; Bath and Eddy 1979) and may be explained by differential changes in red cell and plasma volumes which appear to be species specific.

The results of the present laboratory experiment indicated that salinity had a negative effect on the growth of all three species, particularly at salinities above the isotonic level. These findings are in agreement with Shaw et al. (1975), Clarke et al. (1981), McKay and Gjerde (1985), and McCormick et al. (1989c), who also found that isotonic salinity was not the point

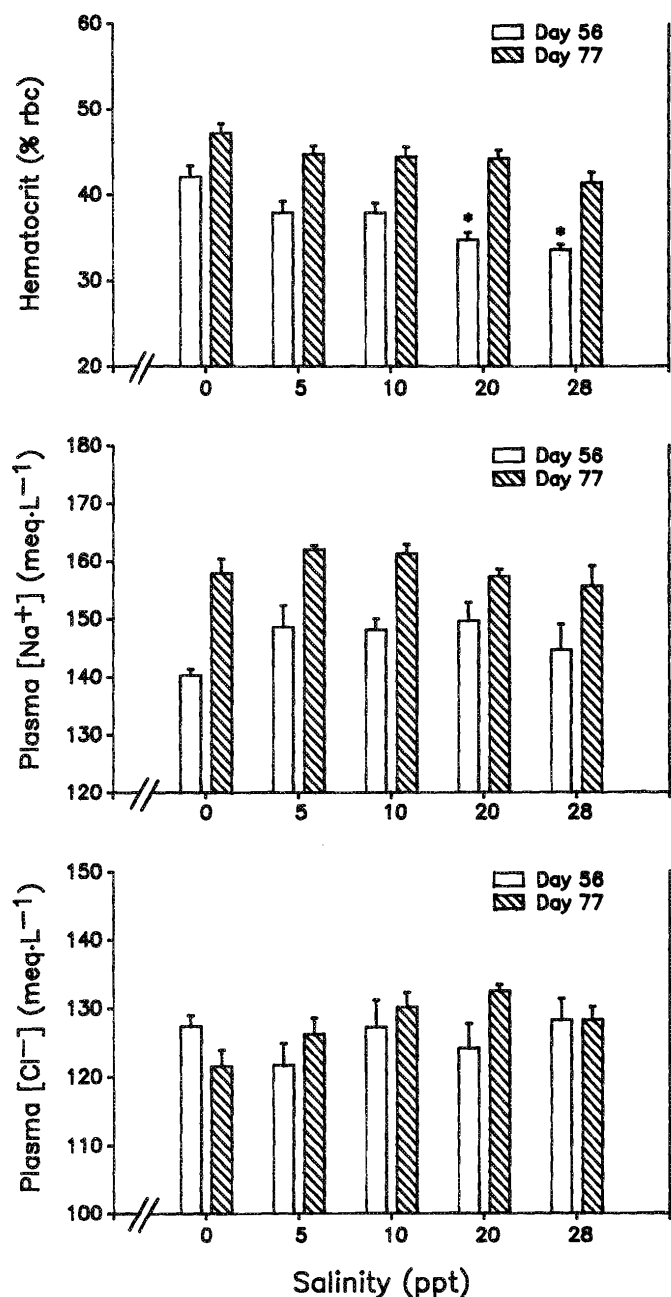


FIG. 5. Mean hematocrit and $[Na^+]_{pl}$ and $[Cl^-]_{pl}$ (± 1 SE) for chinook salmon fry reared in five different salinities. Asterisks denote salinity treatment means which are significantly different from the freshwater value.

of maximal growth in juvenile salmonids. The works of Canagaratnam (1959) and Otto (1971) with coho salmon presmolts are widely quoted in the literature as studies which support the notion of an optimum growth response in an isotonic environment. Subsequent experiments, however, have revealed conflicting results. Canagaratnam (1959) used small aquaria with static water to study the growth response of sockeye and chum (*Oncorhynchus keta*) as well as coho salmon fry to different salinities. This resulted in low overall specific growth rates (1.25%·d⁻¹ for coho fry in fresh water) and very high mortalities in the sockeye and chum fry tests. It is possible, therefore, that confinement stress affected the study results. In the exper-

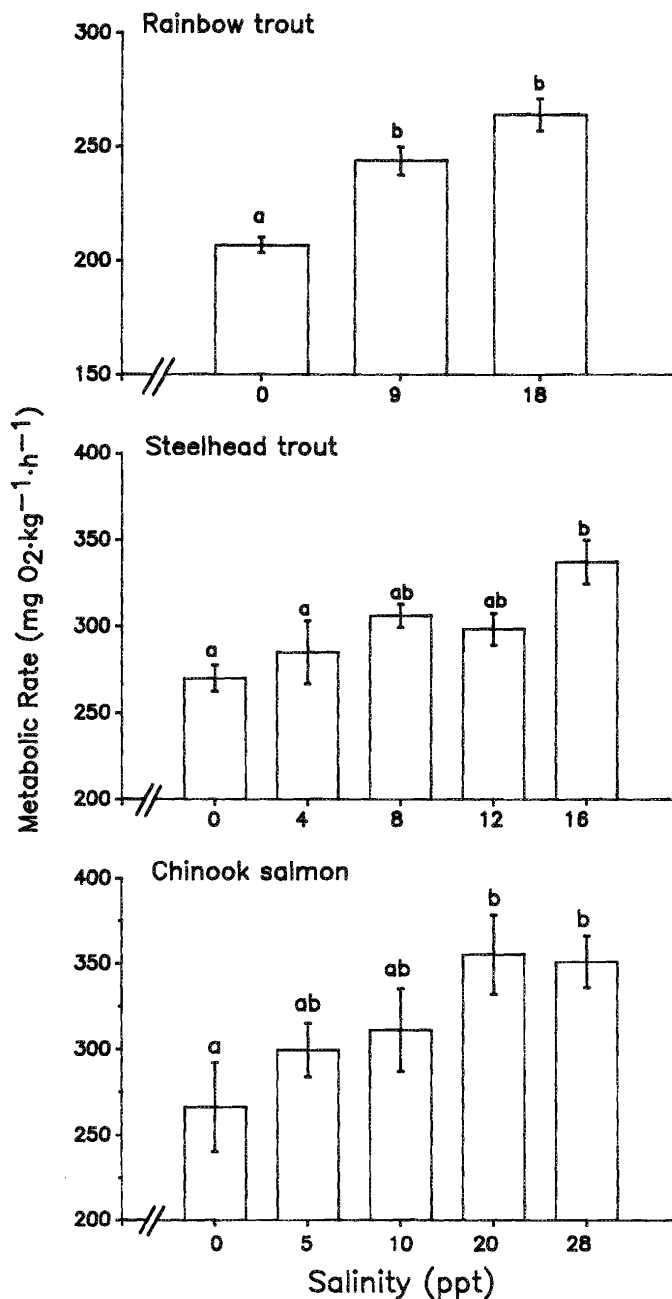


FIG. 6. Metabolic rates of juvenile rainbow and steelhead trout and chinook salmon in different salinities. Means (± 1 SE) with a common letter are not significantly different by Tukey's test.

iments conducted by Otto (1971), growth rates of coho salmon fry were actually higher in fresh water from June to September and increased in 10 ppt salinity only from October to February. This shift in salinity for optimal growth during the fall is consistent with other studies on the behaviour and physiology of coho salmon presmolts. Otto and McInerney (1970) found that the preferred salinity for the same stock of coho salmon fry changed during October and November to 10 ppt, corresponding with the growth results. Conte et al. (1966) also found that juvenile coho salmon were able to tolerate full-strength seawater in October, 6 mo prior to the usual seawater migration period. Lasserre et al. (1978) further showed that the activity of gill Na^+, K^+ -ATPase, the enzyme located in chloride cells

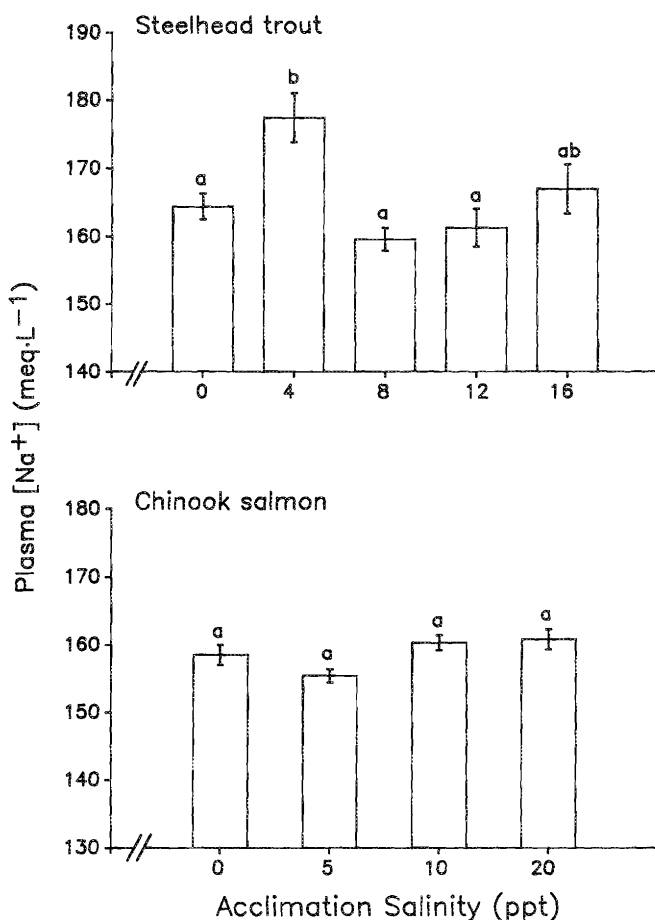


FIG. 7. $[Na^+]_p$ of juvenile steelhead trout and chinook salmon following a 24-h seawater challenge test. Means (± 1 SE) with a common letter are not significantly different by Tukey's test.

which is indicative of ion excretion capacity (see review by Folmar and Dickhoff 1980), increased during mid-October in coho salmon underyearlings in fresh water. This resulted in improved growth and survival of the fish when they were transferred to seawater netpens during autumn compared with other times of the year (Harache et al. 1980). These results are all consistent with the observation that coho salmon are sometimes found in estuaries as underyearlings (Tschaaplinski 1987) and have therefore evolved physiological mechanisms to optimize growth in changing environments. A similar rise in gill Na^+ , K^+ -ATPase activity and estuarine residence during autumn has also been reported for Atlantic salmon parr (Langdon and Thorpe 1985; Cunjak et al. 1989). Determinations of the optimal salinity for growth in salmonids must therefore consider the influences of species-specific differences, developmental stage, size, and season (Saunders and Henderson 1969b; McCormick and Naiman 1984; McCormick et al. 1989c).

The highest growth rates of juvenile salmonids in the present study were associated with fresh water, which is also the natural habitat for these species at this life stage. Growth of landlocked rainbow trout fry decreased 53% from fresh water to the hypertonic salinity of 18 ppt, while the growth of anadromous steelhead trout and chinook salmon fry declined only 16 and 12%, respectively, over a similar range. These data indicate that even at the fry stage, anadromous steelhead trout and chinook salmon adapt and grow better at hypertonic salinities than the fresh-

water resident rainbow trout. Although not examined in the present study, it is possible that the steelhead trout and chinook salmon fry possessed a greater number of, or more active, ion-excreting chloride cells on the gill, which are known to be poorly developed and fewer in number in strictly freshwater fish than in anadromous or marine species (Folmar and Dickhoff 1980; Burton and Idler 1984).

Contrary to our initial hypothesis, we found that oxygen consumption rates for all three species of salmonids were not lowest in isotonic salinity; they were lowest in fresh water. The higher metabolic rates at higher water salinities apparently reflected a significant energetic cost, as growth rates declined with increasing salinity, and correlated very well with the changes in oxygen consumption. Assuming that these increased energy demands were related solely to ionic and osmotic regulation (= ion-osmotic regulation), the metabolic rate data suggested that the energetic cost of ion-osmotic regulation for fry of these salmonid species was lowest in fresh water and increased to 12–18% in isotonic salinity.

It is difficult to reconcile the differences in the metabolic responses to changes in salinity found in the present study with those previously reported. In other studies, total metabolic rates measured in acclimated fish have either increased, decreased, or remained unchanged as salinity increased. From the recent literature (Table 4), we have compiled and categorized five patterns of metabolic responses of teleost fish to altered salinities, some of which are similar to those proposed by Nordlie (1978). These patterns include the following: (I) metabolic rate does not change over a wide range of salinities, (II) metabolic rate is minimum in isotonic salinity and increases at lower and higher salinities, (III) metabolic rates are minimum in fresh water and increase at higher salinities, (IV) metabolic rates are highest in fresh water and decrease to isotonic (higher salinities are not tolerated), and (V) metabolic rates are lowest in seawater and increase in lower salinities.

Life habits will, to a certain extent, determine the type of metabolic response to changes in salinity (Table 4). Euryhaline fish which move freely through waters of varying salinity generally demonstrate a type I response, i.e. metabolic rates do not change. Euryhaline forms which are found in fresh water as adults or in estuaries as juveniles tend to illustrate a type II metabolic response (minimum in isotonic salinity) whereas fish which are found as juveniles in fresh water or in estuaries as adults display a type III response pattern (metabolism increases with salinity). Stenohaline freshwater fish show a type IV response, which probably reflects a depression of metabolic rate due to stress. Finally, fish species which are found primarily in marine habitats illustrate a type V response, which is in accordance with Prosser's (1973) general hypothesis that for euryhaline marine species, metabolic rates increase at reduced salinities. It is also apparent from this summary of metabolic patterns that juveniles frequently respond differently than do adults of the same species (e.g. *O. tshawytscha*, *Ambassis interrupta*) and, in general, are more influenced by salinity variations than are the larger adults, a point also recognized by Nordlie (1978). It is evident from the studies cited in Table 4, therefore, that the lowest metabolic rates in response to salinity are associated with the environments in which the species are most commonly found, and presumably most physiologically adapted, for a particular life stage. The strong correlation between optimum growth and metabolic rates in response to salinity observed in the present study supports this hypothesis.

TABLE 4. Metabolic response patterns of teleost fish to changes in environmental salinity, compiled from recent literature.

Metabolic response	Species and life stage	Habits ^a	Reference
I. No significant change in metabolic rate over a wide salinity range	<i>Oncorhynchus tshawytscha</i> (chinook salmon), subadult	Anad-Est/Mar	Bullivant 1961
	<i>Kuhlia sandvicensis</i> (aholehole), adult	Eur-Mar	Muir and Niimi 1972
	<i>Aphanius dispar</i> (cyprinodont), juvenile/adult	Eur-Mar	Shakhauge and Lotan 1974
	<i>Gillichthys mirabilis</i> (gobie), adult	Eur-Mar	Courtois 1976
	<i>Ambassis interrupta</i> , adult	Eur-Est	Nordlie 1978
	<i>Fundulus heteroclitus</i> (mummichog), adult	Eur-Bra	Nead and Buttner 1987
	<i>Girella nigricans</i> (percoid), adult	Eur-Mar	Fang 1982
II. Metabolic rate is minimum in isotonic salinity and increases at lower and higher salinities	<i>Oncorhynchus mykiss</i> (rainbow trout), adult	Eur-Fw	Rao 1968
	<i>Tilapia nilotica</i> (tilapia), adult	Eur-Fw	Farmer and Beamish 1969
	<i>Brevoortia tyrannus</i> (Atlantic menhaden), juvenile	Eur-Est	Hettler 1976
	<i>Ambassis interrupta</i> , juvenile	Eur-Est	Nordlie 1978
	<i>Mugil curema</i> (white mullet), juvenile	Eur-Est	Fanta-Feofiloff et al. 1986
	<i>Oreochromis niloticus</i> (tilapia), juvenile	Eur-Est	DeSilva et al. 1986
	<i>Oreochromis mossambicus</i> (tilapia), adult	Eur-Fw	Febry and Lutz 1987
III. Metabolic rate is lowest in fresh water and increases with salinity	<i>Platichthys stellatus</i> (starry flounder), juvenile/adult	Eur-Est	Hickman 1959
	<i>Mugil cephalus</i> (striped mullet), adult	Eur-Est/Mar	Nordlie and Leffler 1975
	<i>Cyprinodon salinus</i> (pupfish), adult	Eur-Bra	Stuenkel and Hillyard 1981
	<i>Oncorhynchus kisutch</i> (coho salmon), juvenile	Anad-Fw	Nagibina 1983
	<i>Leiostomus xanthurus</i> (spot), juvenile	Eur-Est	Moser and Hettler 1989
	<i>Oncorhynchus mykiss</i> (rainbow and steelhead trout), juveniles	Anad-Fw	This study
	<i>Oncorhynchus tshawytscha</i> (chinook salmon), juveniles		
IV. Metabolic rate is highest in fresh water and decreases to isotonic (higher salinities are not tolerated)	<i>Ctenopharyngodon idella</i> (grass carp), juvenile	Sten-Fw	Maceina et al. 1980
	<i>Ictalurus nebulosus</i> and <i>I. punctatus</i> (catfish), adult	Sten-Fw	Furspan et al. 1984
V. Metabolic rate is minimum in seawater and increases in lower salinities	<i>Mugil cephalus</i> (striped mullet), adult	Eur-Mar	Marais 1978
	<i>Mugil macrolepis</i> (mullet), adult	Eur-Mar	Mathew 1978
	<i>Cyprinodon variegatus</i> (sheepshead minnow), juvenile/adult	Eur-Bra	Barton and Barton 1987

^aAnad, anadromous; Est, estuarine; Mar, marine; Eur, euryhaline; Bra, brackish; Fw, fresh water; Sten, stenohaline.

Given that the mechanisms of ion-osmotic regulation are similar among fish species (Evans 1984), the variety of metabolic responses to different salinities depending on species, habits, and life stages suggests that these responses are being influenced by physiological processes in addition to the energy required for ion-osmotic regulation (Febry and Lutz 1987).

McCormick et al. (1989a), for example, have recently reported that the changes in the metabolic capacity of isolated gill and kidney tissues of Atlantic salmon in different salinities did not account for the magnitude of changes seen in whole-body metabolism. During acclimation to different salinities, hormonal changes occur in fish which, in addition to their role in

ionic regulation, may also have secondary effects on metabolism not directly related to ion exchange between blood and water. Cortisol, for example, is known to promote branchial ion excretion by stimulating chloride cell proliferation, differentiation, and secretory activity (Foskett et al. 1983; McCormick et al. 1989b; Madsen 1990a, 1990b) and shows peaks during the parr-smolt transformation of juvenile salmonids in fresh water (Specker and Schreck 1982). This hormone has also shown long-term increases in juvenile salmonids after entry into seawater (Redding et al. 1984; Young et al. 1989; Avella et al. 1990) and has been associated with increases in metabolic rate (Chan and Woo 1978; Barton and Schreck 1987). Therefore, the increase in metabolic rates observed for juvenile salmonids in the present study may also have been related to a more general metabolic response of the test species to increasing salinity, as opposed to a strictly energetic cost of ion-osmoregulation. The confounding effects of salinity changes on overall metabolism make it difficult to distinguish metabolic costs exclusively associated with ion-osmotic regulation. The estimates provided previously, therefore, probably overestimate the energetic cost of ion-osmotic regulation. More detailed and quantitative descriptions of the physiological responses of fish to changes in water salinity that account for differences in species, habits, and life stages are necessary to accurately assess the energetic cost of ionic regulation. For example, comparison of gill oxygen consumption with total body metabolism in vivo may assist in separating energetic activities associated with ionic regulation from other metabolically active processes involved with adjustments to alterations in water salinity.

Acclimation to dilute salinities did not improve the osmoregulatory performance of juvenile steelhead trout and chinook salmon challenged with an acute (24-h) exposure to full-strength seawater. Similar results were obtained by Clarke et al. (1981) for coho and chinook salmon reared at 10°C. The use of this technique in a hatchery prior to seawater transfer is therefore probably not necessary, once the fish have reached the critical size and developmental state for smolting. Freshwater rearing temperature, fish size, and time of transfer have been shown to be the most important factors influencing adaptability and growth in seawater for juvenile fall chinook salmon (Clarke and Shelbourn 1985). The absence of mortalities during the 24-h seawater challenge tests were unexpected for the 7- to 9-g steelhead trout, whose critical size for seawater adaptation is generally accepted to be about 35 g (Conte and Wagner 1965; Clarke 1982). A possible explanation for these findings may be that a constant spring photoperiod (14 h) was used throughout the steelhead experiment, and this has been shown to accelerate seawater adaptation in juvenile steelhead trout (Johnsson and Clarke 1988).

In summary, growth and metabolic rates of rainbow trout, steelhead trout, and chinook salmon fry were optimum in fresh water, their natural habitat at this life stage. Isotonic salinity did not provide metabolic or growth advantages in the present study, despite the hypothesis that it would provide the lowest energetic demands for ion-osmoregulation. Comparison with other studies indicated that optimal salinities for growth and metabolic rates were influenced by species, life stage, and season. Although the oxygen consumption data suggested that the energetic cost of ion-osmotic regulation increased with salinity, attempts to quantify this cost were probably affected by other metabolic processes which respond to changes in salinity.

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